

Econ 30065: Microeconomics

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Profit Maximization

Main Topics

- Profits, revenues and costs
- Marginal revenue, marginal cost, and profit maximization
- Supply decisions by price-taking firms

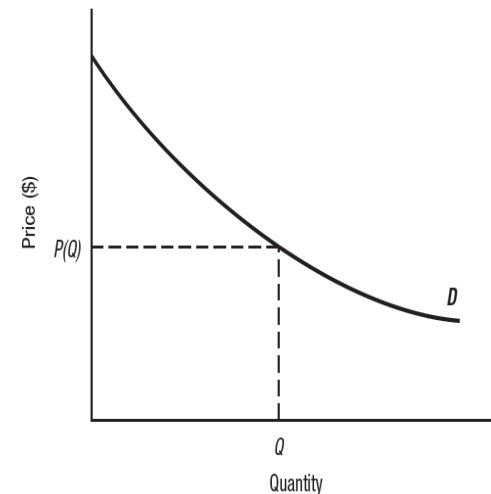
Maximizing profits

- Profits: firm's revenue minus its costs:

$$\Pi = TR - TC$$

- **Revenues: $TR = P(Q) \times Q$**

- In general: If a firm wants to sell Q units, it needs to set a price $P(Q)$ from inverse demand function:

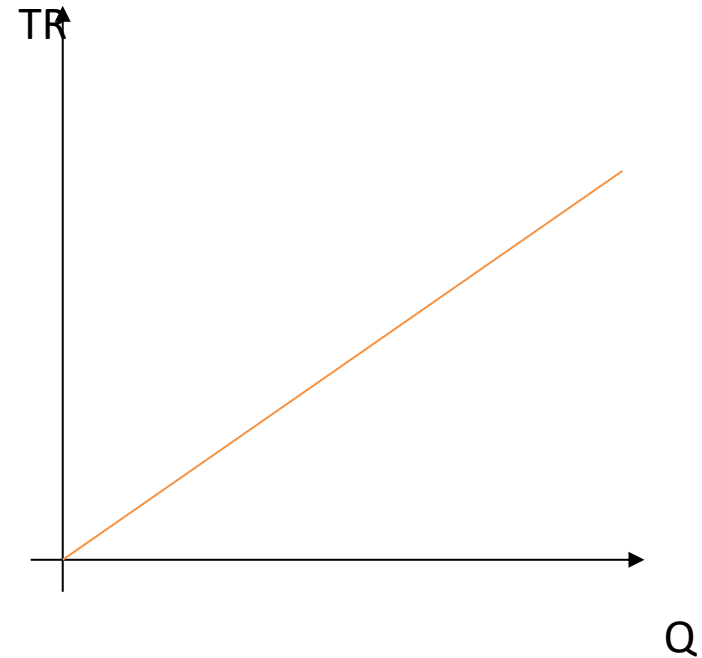
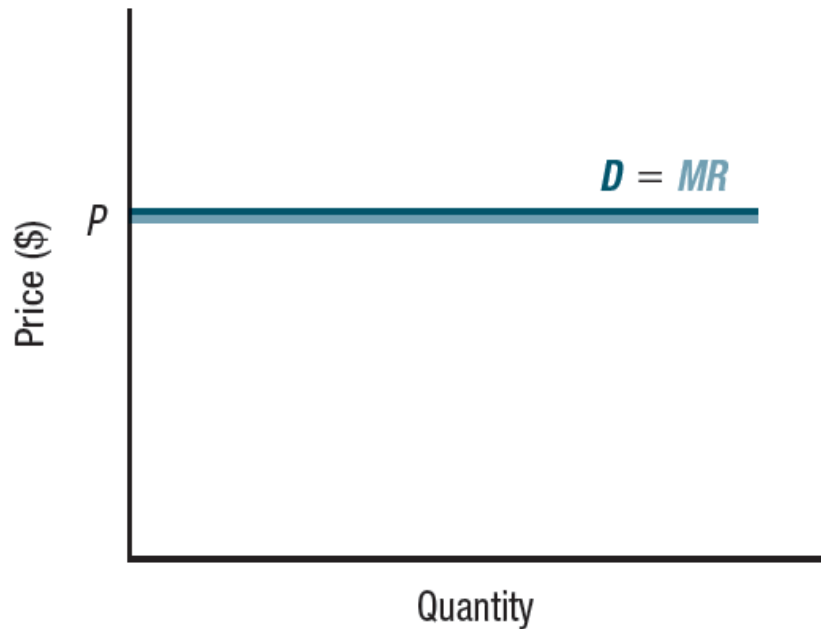


Finding Profit-Maximizing Sales Quantity for Price-Taking Firms

- In perfectly competitive markets price does not depend on single output decisions by a seller: a seller cannot change the market price through his choice of Q .
- **Price-taking firm:** A firm that can sell as much as it wants at P , but nothing at higher/lower price
 - Face a perfectly horizontal demand curve

Finding Profit-Maximizing Sales Quantity for Price-Taking Firms

(a) Horizontal demand curve



Maximizing profits in perfect competition

- $TR = \underline{P} \times Q$
- **Costs:** $TC(Q)$ (from past class cost minimization)
- Thus, for a given price $\underline{P}$ in perfect competition, to maximize profits a firm chooses a sales output Q to maximize revenues minus costs:

$$\text{Max } \Pi = P \times Q - TC(Q)$$

Profit maximization

- How to (mathematically) maximize a function?
 - Take the derivative of the function and set it =0
- $\text{Max } \Pi = \text{TR}(Q) - \text{TC}(Q)$
 - Marginal Revenue – Marginal cost = 0
 - **Marginal Revenue = Marginal cost**

Marginal Revenue

- **Marginal revenue:** additional revenue produced by an increase in the output sold

$$MR = \partial TR / \partial Q$$

Economic intuition for profit maximization

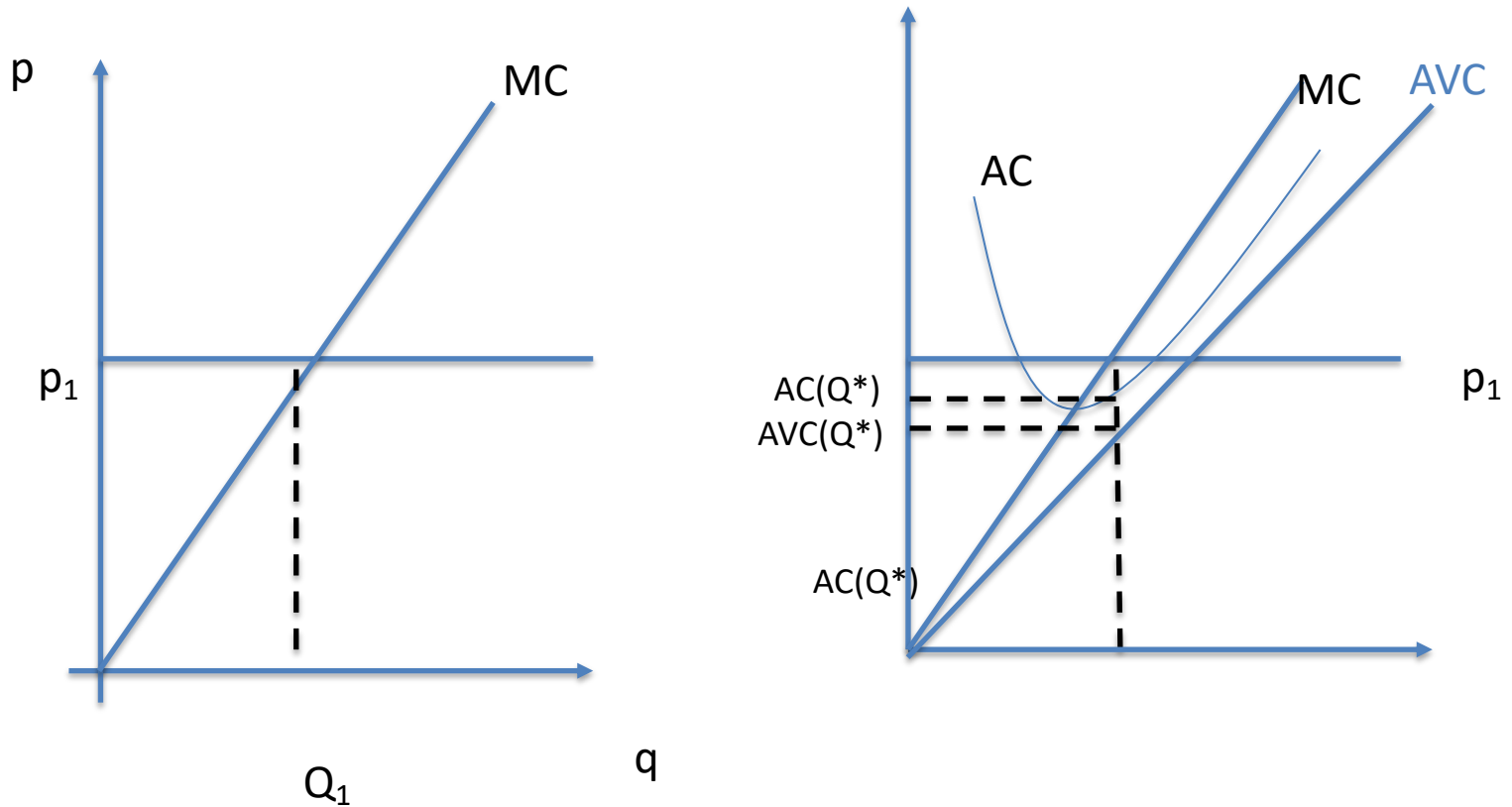
$$\mathbf{MR=MC}$$

- If $MR > MC \rightarrow$ profits go up if firm sells more
- If $MR < MC \rightarrow$ profits go down if firms sells more
- Optimal to choose Q such that $MR=MC$

Short run Finding Profit-Maximizing Sales Quantity for Price-Taking Firms

- Step 1: **Quantity rule.** For a price-taking firm, $P = MR$ for all quantities. Identify positive sales quantities where $P = MC$. If it is satisfied by more than one positive sale then determine which produces the highest profit.
- Step 2: **Shut-down rule.** Check whether the most profitable positive sale quantity results in greater profit than shutting down. Alternative ways:
 - check whether revenues are greater than variable costs:
 $P \times Q^* \geq VC(Q^*)$
 - $P \geq AVC(Q^*)$

Short run Finding Profit-Maximizing Sales Quantity for Price-Taking Firms



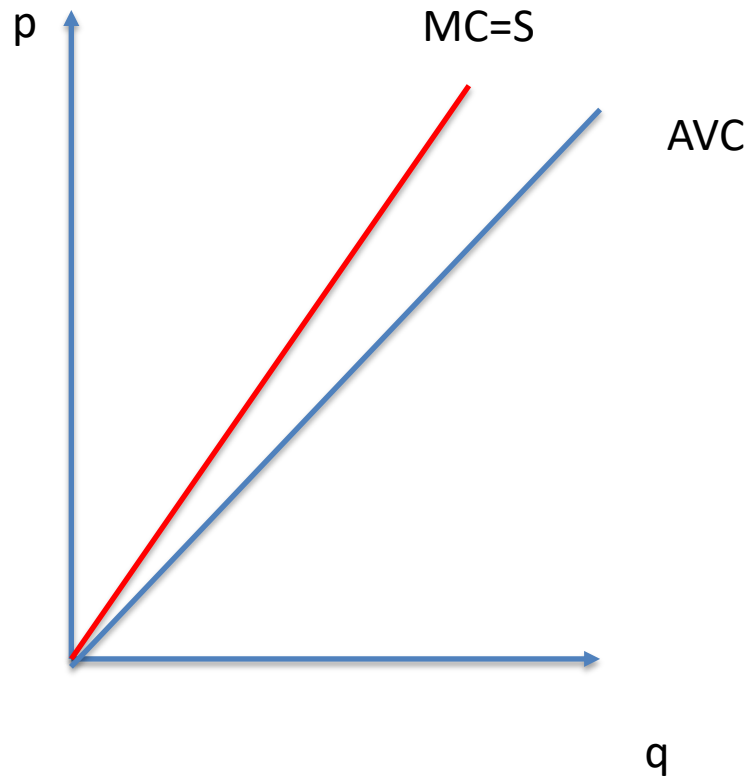
Short run Supply Function of a Price-Taking Firm

- **Supply function:** how many units a firm wants to sell at each possible price P :

$$\text{Quantity Supplied} = S(P)$$

- From profit maximization, at each price P a firm wants to sell Q such that $P = MC(Q)$, and $Q = 0$ for $P < AVC_{\min}$

Short run Supply Curve of a Price-Taking Firm



Summing up

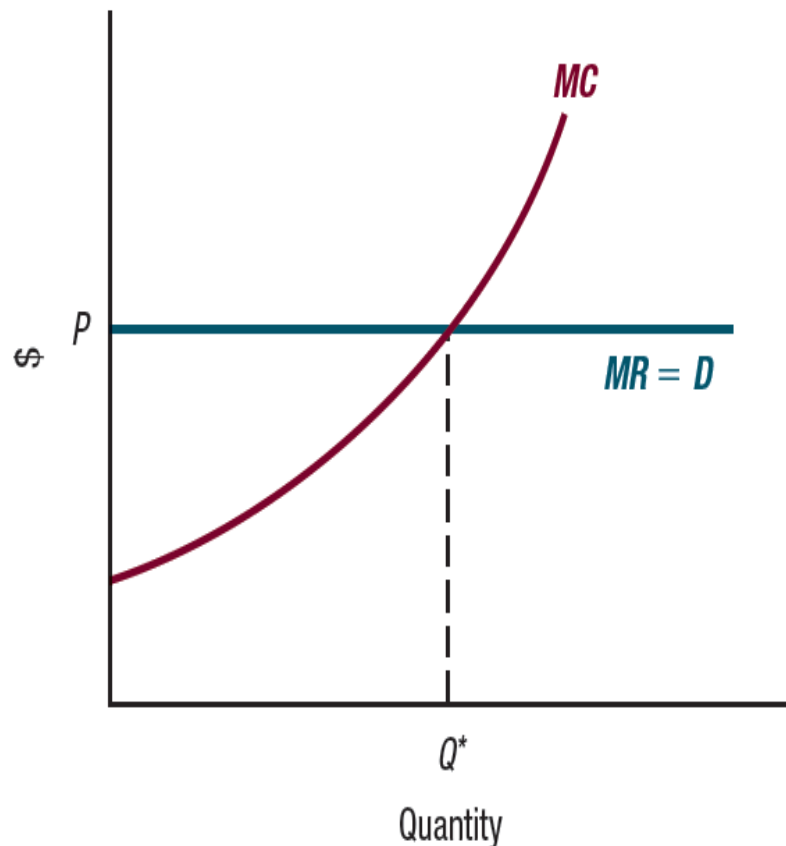
- Increase in price of a variable input that raises the per unit cost of production
 - AVC, MC curves shift up
 - Supply curve shifts up
- Increase in a fixed cost
 - AVC unaffected
 - MC unaffected
 - Supply curve does not shift

Long run: Finding Profit-Maximizing Sales Quantity for Price-Taking Firms

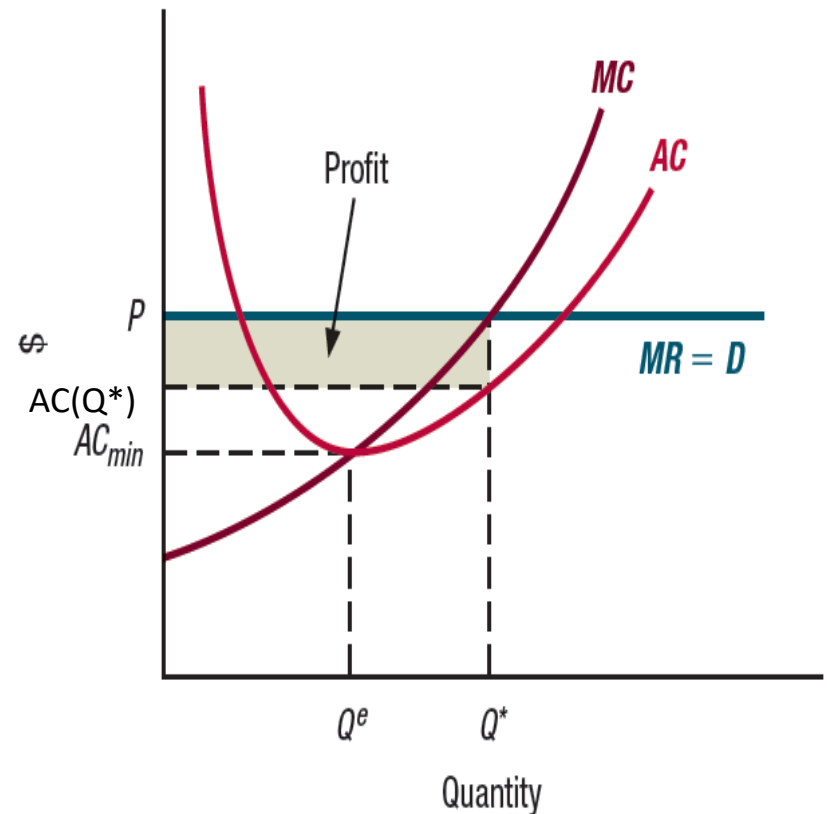
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 - check whether revenues are greater than variable costs:
 $P \times Q^* \geq TC(Q^*)$
 - $P \geq AC(Q^*)$

Long run: Finding Profit-Maximizing Sales Quantity for Price-Taking Firms

(a) The quantity rule



(b) The shut-down rule



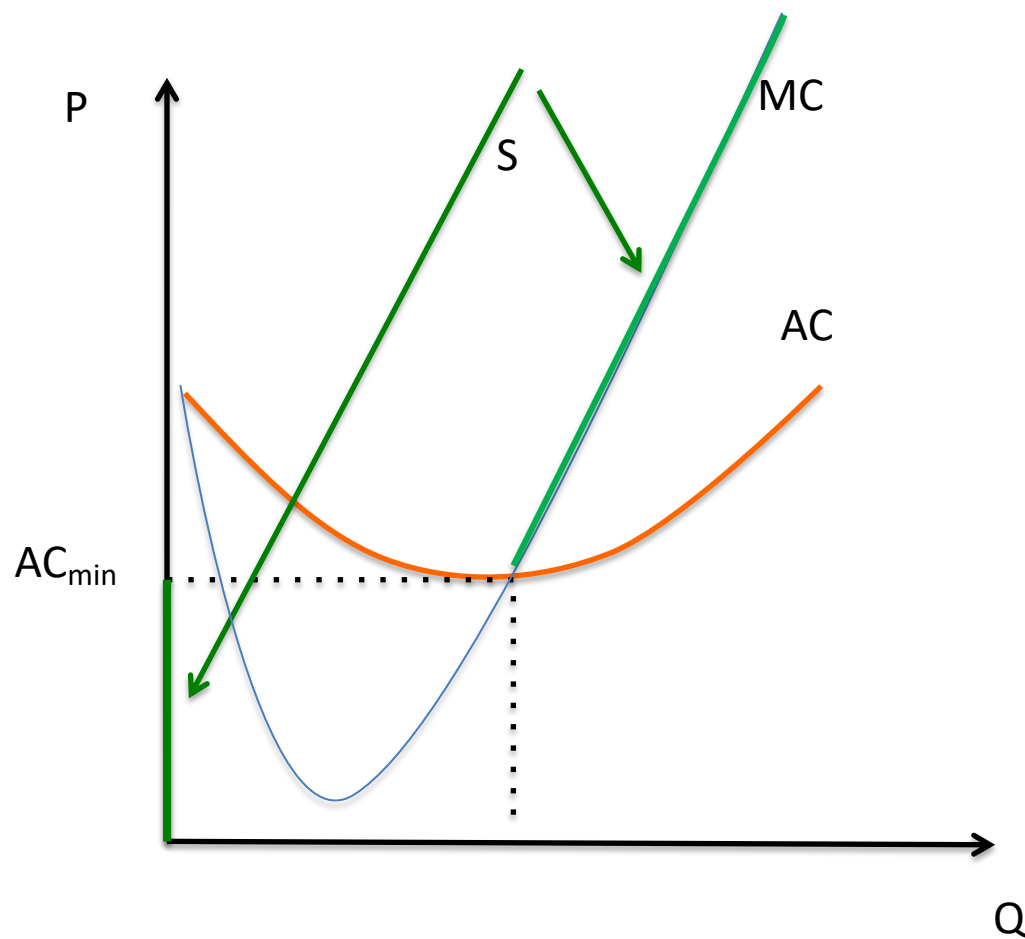
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Supply Curve of a Price-Taking Firm



Law of Supply

- When market price increases, the profit-maximizing sales quantity for a price-taking firm never decreases (i.e. supply is upward sloping)